

Problem

Find the rectangular form of the following complex numbers:

(a) $12\angle -60^\circ$, (b) $-50\angle 285^\circ$, (c) $8e^{j10^\circ}$, (d) $20e^{-j\pi/3}$.

Step-by-step solution

Step 1 of 4

(a)

Consider the complex number, $-8\angle 210^\circ$.

Convert the complex number into rectangular form.

$$\begin{aligned} -8\angle 210^\circ &= -8\cos(210^\circ) - j8\sin(210^\circ) \\ &= -8(-0.866) - j8(-0.5) \\ &= 6.928 + j4 \end{aligned}$$

Thus, the complex number $-8\angle 210^\circ$ in rectangular form is $6.928 + j4$.

Step 2 of 4

(b)

Consider the complex number, $40\angle 305^\circ$.

Convert the complex number into rectangular form.

$$\begin{aligned} 40\angle 305^\circ &= 40\cos(305^\circ) + j40\sin(305^\circ) \\ &= 40(0.5735) + j40(-0.8191) \\ &= 22.94 - j32.77 \end{aligned}$$

Thus, the complex number $40\angle 305^\circ$ in rectangular form is $22.94 - j32.77$.

Step 3 of 4

(c)

Consider the complex number, $10e^{-j30^\circ}$.

Convert the complex number into rectangular form.

$$\begin{aligned} 10e^{-j30^\circ} &= 10\cos(-30^\circ) + j10\sin(-30^\circ) \\ &= 10(0.866) + j10(-0.5) \\ &= 8.66 - j5 \end{aligned}$$

Thus, the complex number $10e^{-j30^\circ}$ in rectangular form is $8.66 - j5$.

Step 4 of 4

(d)

Consider the complex number, $50e^{j\frac{\pi}{2}}$.

Convert the complex number into rectangular form.

$$\begin{aligned}50e^{j\frac{\pi}{2}} &= 50\cos\left(\frac{\pi}{2}\right) + j50\sin\left(\frac{\pi}{2}\right) \\&= 50(0) + j50(1) \\&= j50\end{aligned}$$

Thus, the complex number $50e^{j\frac{\pi}{2}}$ in rectangular form is $\boxed{j50}$.